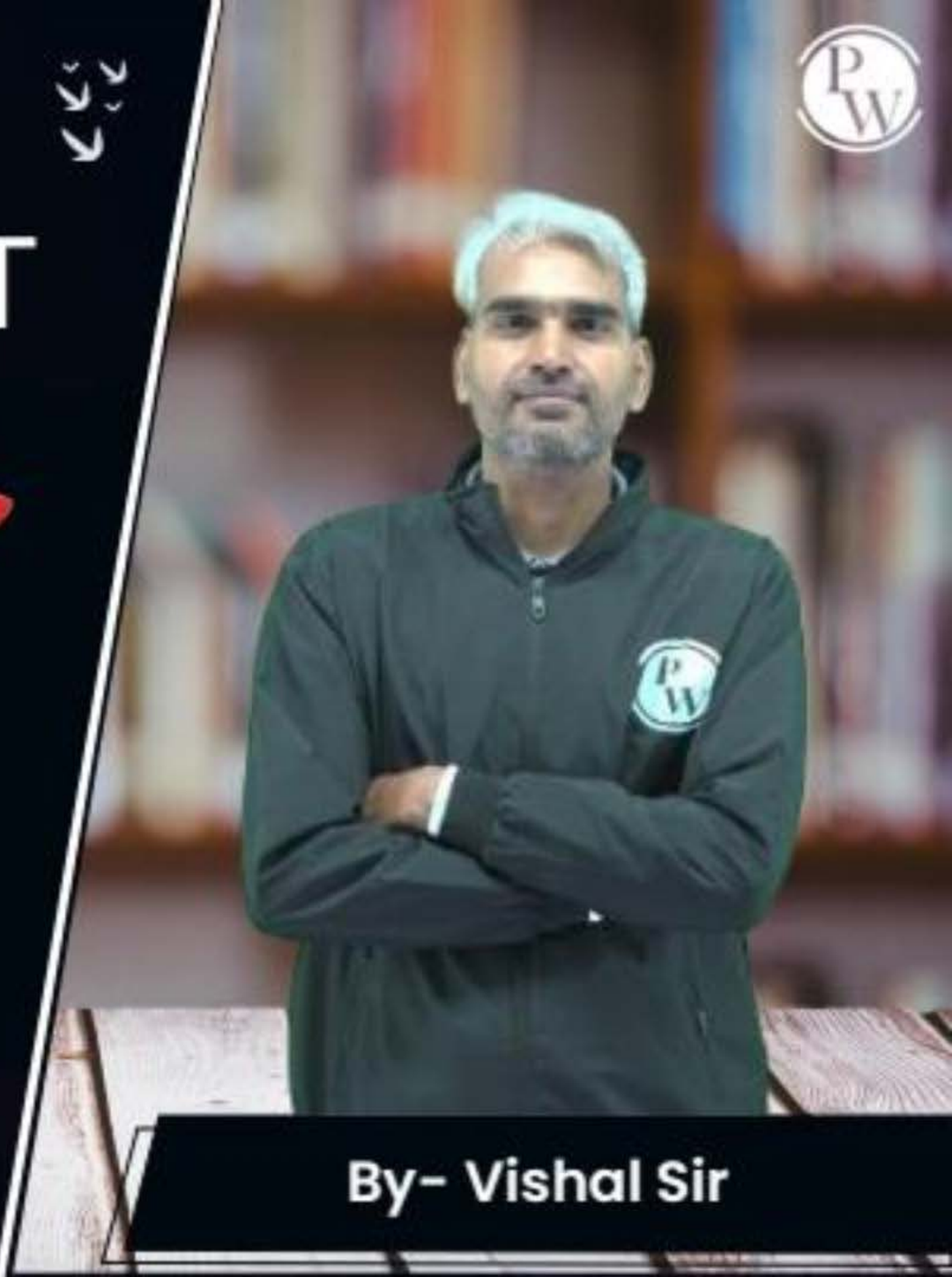
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 01



By- Vishal Sir

Topics to be Covered



- Topic: Syllabus
- Topic: Introduction to DBMS
- Topic: Relational Database Model

Slide



Topic : Syllabus

Weightage : 7 - 10

- Relational Model & Normal Forms
- Query Languages
  - Relational Algebra (Imp)
  - SQL (Most imp.)
  - Tuple Relational Calculus
- Transactions and Concurrency Control
- File Organization & Indexing
- ER model



Topic : Introduction to DBMS

Database is organized collection of related information

Data → {raw facts} → May not provide any information

{10, 20, 35, 60}

Information → {Data with meaning}

10 - No. of students in Vijay batch
20 - No. of students in Parakram batch
35 - No. of students in Shreshtha batch
60 - No. of students in Super60 batch

Database

DBMS : Database management system is a software used to manage and access the database efficiently.

Information can be organized in multiple ways, so different database models are used

eg: ER model
Network model
Hierarchical model
Relational model
object oriented model
etc.

In this course our main focus will be on relational model

In relational model information is organized in the form of table

→ If we store the information gathered in the previous slide in the form of table.

i.e.:

Batch_name	No-of-students
Vijay	10
Parakram	20
Shreshtha	35
Super60	60

And it satisfies the constraints specified by relational model, then it can be called a relation

- "File system" can be used to manage and access the database, but file system fails to do the task efficiently if database is too large.
- Database files are stored in the non-volatile memory i.e; secondary memory.
- Unit of transfer between secondary memory and main memory is "one disk block".

✓ **IO Cost:-** IO cost of a "record access" can be defined as number of secondary memory disk blocks that needs to be transferred from secondary memory to main memory in order to access that record from the database

{ it is stored in the blocks of the disk }

Student.txt file

"other information related to Student"

	Sid	Sname	- - - - - -
B_1	S_1 S_{100}		
B_2	S_{150} S_4		
B_3	S_{227} S_{15}		
B_4	S_{225} S_3		
	⋮	⋮	⋮
B_N	S_2 S_{11}		

Let 'N' blocks of the disk are required to store the file Student.txt

(Too large)

* Retrieve record of the student with Sid = S_{240}

{ Manually Search for record }

↓

It will be time Consuming

{ Write a program to search for Sid = S_{240} }

↓

To write a program we need to know the Physical storage details of the file.

{ it may be a security issue }

{ it is stored in the blocks of the disk }

Student.txt file — "other information related to Student"

	Sid	Sname	- - - - - -
B_1	S_1 … S_{100}		
B_2	S_{150} … S_4		
B_3	S_{227} … S_{15}		
B_4	S_{225} … S_3		
	⋮	⋮	⋮
B_N (Too large)	S_2 … S_{11}		

Let 'N' blocks of the disk are required to store the file Student.txt

- Retrieve record of the student with Sid = S_{240}

Even if we have written a program to compare the existing Sids with "S_{240}", then we will have to transfer the blocks of file Student.txt from secondary memory to main memory (one by one) until the block containing the record with Sid = S_{240} is found.

In worst case we may have to transfer all the blocks of the file from secondary memory to main memory & it will be time consuming.

i.e; with file system IO cost may be too high.



2 mins Summary

Topic Introduction to DBMS

Topic Relational database

Slide

THANK - YOU